

The Indonesia Nickel Lock: How One Nation's Export Ban Is Redrawing the Global EV Battery Map

Blue Lotus Research Intelligence | CivilisationOS
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Data Sources Register (Section X — presented first for transparency)

All market figures in this report are sourced from live web searches conducted on 2026-08-28. No training-data market figures are used. Citations embedded throughout.

Primary Live Sources:

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I. Opening: The Nation That Said No to the WTO

On 30 November 2022, the World Trade Organization's dispute settlement panel published a ruling that landed on Jakarta's desk like a diplomatic grenade. Indonesia's ban on the export of raw nickel ore, the panel declared, violated the WTO's prohibition on export restrictions. The ruling was unambiguous: Indonesia must dismantle the measure that had been in place since January 2020. The European Union, which brought the complaint on behalf of European steelmakers dependent on Indonesian ore feedstock, considered the ruling a vindication of rules-based trade.

Jakarta read the same ruling and reached a different conclusion. On 12 December 2022, Indonesia filed an appeal. The appeal is almost certain to gather dust. The WTO's Appellate Body — the institution's ultimate arbitration chamber — has been paralysed since 2019, when the United States blocked all judicial appointments to protest what it viewed as overreach. As of mid-2026, twenty-nine cases queue in procedural limbo. Indonesia's appeal will not be heard for years, perhaps a decade. In the meantime, the export ban stands. And Indonesia has built a smelting empire around it.

This is the story of how one nation chose industrial sovereignty over trade compliance. It is a story about nickel — the metal that sits at the electrochemical heart of every high-energy-density lithium-ion battery that powers the global electric vehicle revolution. It is a story about leverage: how a country sitting atop the world's largest reserves of a critical mineral decided to convert raw geological luck into industrial capability, using the blunt instrument of an export ban as its primary policy tool.

The consequences are radiating outward. China's battery giants have poured tens of billions of dollars into Indonesian processing plants. Korean manufacturers have commissioned their first local battery cell factories. Chinese-built industrial parks now dominate coastlines in Sulawesi and Halmahera. The United States is scrambling to design trade rules that simultaneously reward

Indonesia's supply and penalise China's ownership of it — a logical impossibility that has produced the FEOC paradox: a situation where Indonesian nickel is simultaneously essential to US clean energy ambitions and largely ineligible to support them under current law.

And deep in the laterite soil of Morowali, beneath the acid fog of HPAL plants and the diesel exhaust of nickel pig iron smelters, communities displaced from their fishing grounds are learning what it means to live next to the twenty-first century's most strategically contested commodity.

II. The Nickel Endowment: Geography as Destiny

Indonesia's geological fortune is almost comic in its scale. According to USGS data, the country holds approximately 55 million tonnes of nickel reserves — the largest single national endowment on Earth, representing approximately 42% of global nickel reserves. The nearest competitors — Australia, Russia, and the Philippines — hold materially smaller positions. Indonesia's share of global nickel production is even more lopsided: from a 16% global share in 2017, Indonesia's contribution surged to approximately 54% of global output by 2023, when it produced 2.02 million tonnes. Projections for 2026 range from 60% to 75% of global mined output, depending on quota decisions and production constraints. According to the IEA's Global Critical Minerals Outlook 2026, Indonesia now accounts for roughly half of global mined nickel production, a figure that continues to climb. [Source: Mining Technology, ING Think, IEA via Indonesia Business Post]

This concentration of supply in a single jurisdiction is without precedent in the modern critical minerals era. For comparison, Saudi Arabia's share of global oil production — the most politically weaponised commodity of the twentieth century — peaked at around 13%. Indonesia's position in nickel is five times more dominant.

But raw concentration statistics obscure important technical nuance. Nickel is not a homogeneous commodity. For the purposes of electric vehicle batteries, what matters is the distinction between **Class 1 nickel** (high-purity, minimum 99.8% Ni content) and **Class 2 nickel** (lower purity, primarily used in stainless steel as nickel pig iron or ferronickel). The pathways diverge sharply at the chemistry of the ore body.

Indonesia's deposits are overwhelmingly laterite ores — formed by millions of years of tropical weathering of ultramafic rock. Laterite deposits are abundant and geologically shallow, enabling large-scale open-cast mining. However, they are metallurgically complex. Unlike sulphide ores (which dominate in Canada, Russia, and Australia and can be relatively efficiently processed into Class 1 nickel via conventional smelting and refining), laterite ores require energy-intensive processing to yield battery-grade material.

For electric vehicle batteries, the critical chemistries are **NMC (Nickel Manganese Cobalt)** and, increasingly, **NCA (Nickel Cobalt Aluminium)**. Both require high-purity nickel inputs. NMC 811 — a chemistry formula indicating 80% nickel, 10% manganese, 10% cobalt — maximises energy density and is the preferred chemistry for passenger EVs where range is paramount. Chinese manufacturers have also driven down costs in **LFP (Lithium Iron Phosphate)** chemistry, which uses no nickel at all, creating a parallel supply chain that partially insulates some of the battery market from Indonesian supply. However, NMC chemistry dominates in premium segments and long-range applications, and the consensus projection is that NMC demand will grow alongside total EV penetration, keeping nickel essential to the high-performance segment.

The critical technology linking Indonesian laterite to battery-grade nickel is **HPAL — High-Pressure Acid Leaching**. The HPAL process subjects crushed laterite ore to sulfuric acid at temperatures above 250°C and pressures exceeding 40 atmospheres. Under these conditions, nickel and cobalt dissolve into solution, and through a series of precipitation and purification steps, the output is **MHP (Mixed Hydroxide Precipitate)** or **MSP (Mixed Sulphide Precipitate)** — intermediate products that can be further refined into battery-grade nickel sulphate and cobalt sulphate for cathode precursor manufacturing.

HPAL was long considered a difficult and capital-intensive technology with a troubled engineering track record. Projects in Australia, Cuba, and the Pacific historically suffered from corrosion failures, acid management problems, and chronic cost overruns. Chinese companies — particularly Zhejiang Huayou Cobalt and Tsingshan's subsidiaries — broke this pattern by deploying HPAL at scale in Indonesia at costs and timelines that defied Western industry precedent. This technology deployment at scale is, arguably, the most consequential industrial achievement of the 2020s in critical minerals.

III. The Export Ban Strategy: Building a Value Chain from Ore to Ambition

The story of Indonesia's export ban is not linear. It is a story of policy oscillation — of a government that grasped the logic of downstream value capture, tested it, flinched under pressure, and then committed to it permanently.

2009: Indonesia's Law No. 4 on Mineral and Coal Mining establishes the legal framework requiring domestic processing of minerals before export. It is a legal premonition more than an executed policy.

January 2014: The first export ban takes effect. Indonesia prohibits exports of unprocessed nickel ore and bauxite. The stated logic is straightforward: raw ore exports yield minimal domestic value-added. Foreign smelters capture the processing margin. Indonesia should capture it instead.

January 2017: Under pressure from mining companies and facing a short-term revenue cliff, President Joko Widodo partially reverses the 2014 ban. Low-grade nickel ore (below 1.7% Ni content) is permitted for export, subject to export duties. The reversal undermines three years of downstream investment momentum and is later assessed by analysts at the East Asia Forum as a significant policy setback that damaged investor credibility in the downstream industrialisation programme.

January 2020: The Jokowi government reinstates a full export ban on nickel ore with no exemptions and declares it permanent. This time, the commitment is credible — anchored by bilateral investment agreements with Chinese companies that had already committed capital to Indonesian processing capacity. The WTO complaint from the EU follows.

November 2022: The WTO panel rules against Indonesia. Jakarta appeals and continues the ban.

2023-2026: The export ban holds. Indonesia uses it as leverage to attract investment across the full downstream value chain.

The conceptual logic of Indonesia's downstream strategy can be mapped as a linear value chain: **laterite ore → nickel pig iron (NPI) → Mixed Hydroxide Precipitate (MHP) → precursor cathode active material (pCAM) → cathode active material (CAM) → battery cells → battery packs → electric vehicles**. Each step in this chain adds value, creates employment, develops engineering capability, and embeds Indonesia deeper into the global battery supply system.

Indonesia's position in 2020 was essentially the first step in that chain — a raw ore exporter. Its ambition for 2030 is to be present at every step. This is not merely an economic programme; it is an industrial civilisation project.

The downstream thesis rests on a straightforward insight: that the true value in nickel is not in the ore but in what the ore becomes. A tonne of nickel ore might be worth a few dollars. A tonne of battery-grade nickel sulphate is worth multiples of that. A battery pack powering a vehicle is worth multiples again. Indonesia holds the feedstock monopoly. The question is whether it can convert that monopoly into manufacturing sovereignty, or whether it will remain a processing hub for other nations' industries.

The WTO ruling represents the international trade system's verdict that Indonesia's tool is illegitimate. Indonesia's counter-argument — articulated publicly by former Coordinating Minister

Luhut Binsar Pandjaitan and echoed by the Prabowo government — is that resource sovereignty is non-negotiable: nations have the right to add value to their own natural endowments before exporting them, and the WTO rules were designed by industrial powers who had already completed their own industrialisation. "These minerals belong to us," Finance Minister Purbaya Yudhi Sadewa stated on May 14, 2026. [Source: Indonesia Business Post]

IV. The Chinese Investment Pivot: Capital, Technology, and Integration Logic

If Indonesia provided the ore and the policy framework, China provided the capital, the engineering capability, and the supply chain integration logic that turned the export ban strategy into reality. The first wave of Indonesian nickel downstream investment is, by any honest accounting, predominantly a Chinese industrial achievement.

Tsingshan Group is the foundational player. The private Wenzhou conglomerate built its fortune in stainless steel — a product that requires nickel as a primary input. In the late 2010s, Tsingshan's founder Xiang Guangda made a bold bet: that Indonesia's laterite nickel, processed through Rotary Kiln Electric Furnace (RKEF) technology that Tsingshan had mastered, could produce nickel pig iron (NPI) cheaply enough to supply global stainless steel markets and eventually the battery industry. Tsingshan became the anchor investor and industrial developer of the **Indonesia Morowali Industrial Park (IMIP)** in Central Sulawesi — a joint venture between Tsingshan and local firm Bintang Delapan Minerals. IMIP is now the largest nickel industrial park in the world by output. By 2026, it generates 11.5 million tonnes of industrial waste annually (including tailings), employs tens of thousands of workers, and has its own port, power plant, and logistics infrastructure. Tsingshan subsequently co-invested in **Indonesia Weda Bay Industrial Park (IWIP)** in North Maluku — a second mega-park that has become a hub for HPAL operations.

In June 2026, Tsingshan signed an MOU with UNIDO designating both IMIP and IWIP as pilot projects for Eco-Industrial Parks — a move widely seen as an attempt to address mounting ESG criticism while preserving operating licences. However, the same month, Tsingshan asked NPI producers at Weda Bay to curb output to conserve power for aluminium smelting, illustrating the production trade-offs within the park's shared infrastructure. [Source: VOI Indonesia]

Zhejiang Huayou Cobalt represents the most technically sophisticated Chinese investment in Indonesia. Huayou has eight investment and shareholding projects in Indonesia, concentrated in IMIP and IWIP. The flagship HPAL operations include:

- **PT Huayue Nickel Cobalt (Morowali)**: began construction 2020, production 2021, capacity of 60,000 tonnes per annum nickel-in-MHP.
- **PT Huafei Nickel Cobalt (Weda Bay)**: began construction 2021, production testing from June 2023, capacity of 120,000 tonnes per annum nickel-in-MHP.

Beyond its Indonesian processing facilities, Huayou struck the most geopolitically significant deal of the sector in 2023: a final investment agreement with Ford Motor Company and PT Vale Indonesia to develop the **Pomalaa HPAL project** in Southeast Sulawesi, valued at approximately \$4.5 billion, targeting 120,000 tonnes per annum of MHP. The Pomalaa project is scheduled to commence operations in Q4 2026, mechanically targeting completion around August 2026. [Source: Deccan Herald, Mining Technology]

This Ford-Vale-Huayou joint venture represents a structural collision: a US automaker co-investing with a Chinese battery materials company in Indonesian nickel processing — precisely the kind of arrangement that FEOC rules were designed to prevent from qualifying for IRA tax credits. The partnership illustrates the impossible geometry of the supply chain: US manufacturers need to be in Indonesia, but the Chinese technology and capital that makes Indonesia viable creates IRA ineligibility.

CNGR Advanced Material targets the downstream step beyond MHP — the cathode precursor (pCAM). CNGR announced plans for a facility capable of reaching \$10.5 billion in total investment

across three phases, with the first precursor plant at the Setangga Special Economic Zone in South Kalimantan commencing operations in December 2025. CNGR also doubled its nickel matte capacity at IMIP. The pCAM investment represents Indonesia's most ambitious attempt to move up the battery value chain beyond raw processing — if successful, it would embed Indonesia into cathode manufacturing, not merely mineral supply. [Source: The Jakarta Post, Battery-News]

Why Chinese companies dominated the first wave of Indonesian nickel downstream investment comes down to four structural advantages. First, risk appetite: Chinese conglomerates with state backing or deep domestic capital markets could absorb the elevated risk of pioneering HPAL in a new jurisdiction. Second, HPAL technology ownership: Huayou and its peers had accumulated HPAL engineering capability through earlier projects and had significantly reduced capex costs per tonne of capacity. Third, supply chain integration logic: Chinese battery makers needed secure, low-cost nickel for their cathode materials. Owning the mine-to-MHP chain insulated them from spot market exposure. Fourth, speed: Chinese project timelines in Indonesia have consistently beaten Western benchmarks, with HPAL projects coming online in three to four years versus the seven to ten year timelines common in Western-managed resource projects.

The social costs of this speed are discussed in Section VII. But on pure industrial execution metrics, the Chinese deployment in Indonesian nickel downstream processing represents a generational achievement in applied engineering at scale.

V. The Western and Korean Response: Too Late, Too Constrained

Western and Korean engagement with Indonesia's nickel downstream sector arrived later, in smaller volumes, and is now constrained by a regulatory architecture — the US Inflation Reduction Act's Foreign Entity of Concern provisions — that makes much of what has been built in Indonesia structurally ineligible for US market support.

LG Energy Solution (LGES) made the most direct Western-aligned commitment. In a joint venture with Hyundai Motor Group called **HLI Green Power**, LGES and Hyundai began construction of Indonesia's first EV battery cell production plant in 2022 and commenced mass production in 2023, with the facility officially launching in July 2024. Located in Karawang, West Java, the plant operates with a capacity of 10 GWh. A second phase expansion involving an additional \$2 billion investment to expand capacity by 20 GWh is in planning. [Source: EV Magazine, Hyundai Newsroom]

The LG-Hyundai partnership is structurally different from the Chinese investment model. It targets cell manufacturing — a more downstream step — rather than nickel processing. The ore and MHP feedstock for HLI Green Power's cells comes from Indonesian HPAL plants, most of which are Chinese-owned. This creates a supply chain in which a Korean cell manufacturer depends on Chinese-processed nickel — a dependency that the FEOC rules have been designed to make commercially inconvenient for US market access.

Samsung SDI has not made a direct announcement of a dedicated Indonesia facility as of August 2026. However, cathode precursor materials supplied to Samsung SDI are produced by companies including CNGR, which operates in Indonesia. Samsung SDI's indirect exposure to Indonesian nickel is growing through its supply chain, even without a headline factory announcement.

The **IRA FEOC problem** is the defining structural tension of the Western response. The Inflation Reduction Act's clean vehicle tax credits (Section 30D) require, from 2025 onward, that vehicles cannot contain applicable critical minerals extracted, processed, or recycled by a Foreign Entity of Concern. FEOC is defined to include any entity in which Chinese government entities hold 25% or more equity. The critical mineral requirement rising from 60% (2025) to 70% (2026) means that the proportion of battery mineral value sourced from the US or FTA partner countries must reach 70% to qualify. [Source: Fastmarkets, US Department of Energy]

Indonesia does not have a Free Trade Agreement with the United States. On February 20, 2026, the US and Indonesia signed a reciprocal trade agreement covering tariff arrangements and investment access in critical minerals — with US tariffs on Indonesian goods reduced from 32% to

19%. However, as of August 2026, this does not constitute a Qualifying FTA under IRA definitions. [Source: Indonesia Business Post]

The consequence is devastating for US automakers seeking IRA-compliant battery supply chains: the bulk of nickel being produced in Indonesia — processed in Chinese-owned HPAL plants — does not qualify. Most US and Korean battery makers face a binary choice: source from Indonesia and lose IRA credit eligibility, or source from more expensive Western alternatives (Australian or Canadian operations) that have dramatically smaller supply volumes. **Ford's Pomalaa joint venture with Huayou illustrates the bind:** Ford has co-invested with a Chinese company in Indonesian nickel processing that, under current law, likely renders the resulting material FEOC-ineligible. [Source: Fastmarkets, Benchmark Minerals]

The EU response is more nuanced and arguably more strategically dexterous. The European Commission presented proposals for a Comprehensive Economic Partnership Agreement (CEPA) with Indonesia in June 2026, with a view to signature by end-2026. After nearly a decade of stalled talks, EU-Indonesia trade negotiations have gained momentum, with nickel for EV batteries at the heart of the deal. The EU's Critical Raw Materials Act, which identifies strategic raw materials and sets supply diversification targets, explicitly lists nickel. EU-Indonesian alignment provides European battery makers a regulatory pathway that US manufacturers currently lack. [Source: Agence Europe]

VI. Indonesia's EV Ambition: The Bootstrap Logic

Behind the foreign investment headlines is a more ambitious strategic vision: Indonesia does not merely want to supply the world's battery makers. It wants to build its own EV industry, powered by its own batteries, made from its own nickel. This bootstrap logic — process nickel, attract battery factories, attract EV assembly, build domestic EV market — is embedded in Presidential Regulation No. 55/2019 (PR55/2019) and subsequent policy instruments.

The targets are ambitious:

- 140 GWh of battery manufacturing capacity by 2030 [Source: CSIS]
- Deployment of 2 million electric cars and 12 million electric motorcycles by 2030 [Source: ICCT]
- 600,000 EVs produced annually by 2026 [Source: EV Magazine, Hyundai-LG release]

BYD has been the most consequential foreign EV manufacturer to commit to Indonesia. The Chinese automaker acquired a 126-hectare plot in the Subang Smartpolitan industrial estate in West Java and committed IDR 11.7 trillion (approximately US\$721 million) to establish a factory with annual production capacity of 150,000 vehicles. Production is scheduled to have commenced in early 2026. BYD entered the Indonesian market on import strength, selling 11,220 vehicles by end-2024 before local production began. [Source: CnEVPost, Electrive]

Hyundai has pursued a more systematic localisation strategy. The Korean automaker began assembling the Ioniq 5 at its HMMI plant in Cikarang, Bekasi Regency, West Java — the first EV assembly in Indonesia by a major international brand. By 2026, Hyundai was signalling plans to assemble the Ioniq 9, its three-row electric SUV, locally. This strategy is tightly integrated with the HLI Green Power battery cell facility: Hyundai assembles EVs, powered by batteries made by its joint venture with LG, from nickel processed in Indonesian HPAL plants. The vertical integration — at least partially — is Indonesian. [Source: KED Global, VOI Indonesia]

CATL has gone furthest in articulating full value chain integration. Its \$6 billion Indonesia Battery Integration Project encompasses the full chain from nickel mining and processing at East Halmahera (North Maluku) through to battery cell manufacturing at Karawang (West Java). Cell production at the Karawang facility is scheduled to begin in late 2026, with an initial capacity of 6.9 GWh expanding to 15 GWh. The project is a joint investment with Indonesia Battery Corporation (IBC) — the government's battery industry vehicle — and PT Aneka Tambang (ANTAM), the state mining company. It is the most complete realisation of the bootstrap logic yet attempted. In April 2026, Toyota announced a partnership with CATL to produce batteries in Indonesia for export.

[Source: CATL newsroom, ChinaEVHome, CarNewsChina]

Domestic EV development has been more uneven. **Gesits Motor** — acquired by IBC in 2022 — produces electric scooters with ranges of 45-70 km and aims to scale from 40,000 to 100,000 units per year capacity. However, domestic players have struggled with scale and price competitiveness against Chinese OEMs with far greater manufacturing maturity. **Blue Bird**, Indonesia's largest taxi operator, has been adding electric vehicles to its fleet (600 units in 2025) as a bellwether of commercial EV adoption.

Incentive architecture has undergone significant restructuring. Import-based EV incentives — zero import duties, waived luxury tax, VAT relief for completely built-up imported EVs — expired on December 31, 2025. From January 1, 2026, incentives continue only for EVs produced or assembled locally with at least 40% domestic content, qualifying for a reduced 1% VAT rate (versus the standard 11%). This policy shift is explicit industrial policy: it rewards localisation and penalises imports. A further incentive tranche is expected to be announced in mid-2026 targeting electric motorcycles and lower-cost EVs. [Source: Autoini, IESR]

VII. Environmental and Social Costs: The Price of the Pivot

The speed and scale of Indonesia's nickel industrialisation has generated environmental and social costs that are only beginning to be fully documented — and which represent a structural risk to the "green" credentials of Indonesia's battery supply chain.

HPAL tailings are the most acute environmental challenge. The HPAL process produces large volumes of acidic residue containing sulfuric acid, hexavalent chromium (a known carcinogen), and other heavy metals. The Indonesia Morowali Industrial Park (IMIP) currently generates approximately 11.5 million tonnes of tailings annually — a figure projected to rise as HPAL capacity expands. These tailings are typically stored in engineered containment structures, but containment failures have been documented with increasing frequency. On March 16, 2025, intense flooding caused a tailings containment failure at PT Huayue Nickel Cobalt's facility, impacting 341 households (1,092 people) in Labota Village. A February 18, 2026 landslide inside IMIP killed one employee and suspended operations. Since 2015, over 40 workers have died due to unsafe conditions at IMIP alone. [Source: Business and Human Rights Resource Centre, Mongabay, VOI Indonesia]

Coastal ecosystem destruction in Ganda-Ganda village (Morowali Utara) has been documented through satellite analysis: sediment from mining operations has destroyed coral reefs and degraded fisheries, eliminating the economic foundation of coastal communities that predated the industrial park. A 2026 study published on arXiv established a causal link between nickel processing expansion at IMIP and coastal water clarity degradation in Sulawesi using satellite data. [Source: arXiv 2026]

Carbon intensity is the systemic environmental concern. WRI analysis establishes that for every tonne of nickel produced in Indonesia, an average of 93 tonnes of CO₂ equivalent is released. Indonesia's nickel sector accounted for approximately 22% of national emissions from the energy and industrial sectors in 2023. The majority of RKEF (ferronickel and NPI) operations run on coal power; only one smelter in Indonesia — Vale's Sorowako facility in South Sulawesi — operates primarily on renewable energy, powered by 365 MW of hydropower. IEEFA analysis indicates that 51% of new Indonesian nickel capacity uses high-emission ferronickel processing (approximately 60 tCO₂/tNi), while 49% uses HPAL (lower-emission, but still coal-powered in most cases). [Source: WRI, IEEFA]

The "dirty nickel" controversy has become a live issue for ESG-conscious investors and automakers attempting to market EVs as sustainable products. In April 2026, investors controlling approximately US\$4.5 trillion in assets formally urged automakers to address environmental and human-rights risks tied to nickel sourcing, particularly in Indonesia and the Philippines. The EU's forthcoming battery passport regulations will require supply chain due diligence and carbon intensity disclosure — raising the compliance bar for Indonesian producers. As of August 2026, only

Sorowako can plausibly meet low-carbon standards. Most Indonesian nickel, processed on coal power, will require either decarbonisation investment or face European market access complications.

VIII. Geopolitical Implications: The OPEC of Batteries

Indonesia under President Prabowo Subianto is navigating a geopolitical position of unusual leverage. The country's nickel dominance has elevated it from a mid-tier Southeast Asian economy to a strategic swing state in the competition between the United States and China for control of the clean energy supply chain.

The Prabowo doctrine on nickel combines resource nationalism with tactical non-alignment. On May 14, 2026, Finance Minister Purbaya Yudhi Sadewa declared "These minerals belong to us" — as explicit a statement of resource sovereignty as any nation has made in the post-WTO era. On May 20, 2026, Prabowo told Parliament that exports of strategic commodities would be routed through a state-backed body, beginning with coal and palm oil and eventually extending to all minerals. This would represent a qualitative escalation of state control over commodity flows — a step toward the kind of OPEC-style supply coordination that no mineral-producing nation has previously attempted at this scale. [Source: Indonesia Business Post, Fulcrum]

In February 2026, Prabowo visited Washington D.C. and invited American industry leaders to invest in Indonesian downstream processing — using the nickel near-monopoly as a calling card. The subsequent US-Indonesia trade agreement (February 20, 2026), reducing tariffs from 32% to 19%, was a tangible diplomatic win. But the structural FEOC problem remains: the Chinese ownership architecture of most Indonesian processing capacity means the US cannot simply unlock Indonesian nickel into its clean energy supply chain without either changing its own law or demanding China be bought out of assets it built. [Source: Indonesia Business Post]

China's response has been to deepen the relationship rather than relinquish it. On August 21, 2026, China and Indonesia agreed to boost military ties and work more closely on minerals, energy, and technology. The same week, an Asia Times analysis noted that "Indonesia's nickel nationalism falters in the face of China's tech" — acknowledging that Indonesia's ability to capture the full battery value chain continues to depend on Chinese engineering capability and market access. The Chinese Chamber of Commerce in Indonesia, in May 2026, issued a sharp open letter to Prabowo complaining of arbitrary regulation and corruption — but Chinese investment has not slowed. [Source: China Global South, Asia Times]

The ASEAN centrality doctrine provides Indonesia's conceptual framework for managing these competing pressures. Indonesia frames its position not as alignment with one bloc but as engagement with all — a posture that maximises leverage precisely because both Washington and Beijing need access to Indonesian nickel.

The OPEC comparison is instructive but imperfect. OPEC controls a commodity for which there are no short-term substitutes. Nickel has a partial substitute in LFP battery chemistry (which uses no nickel), but for energy-dense NMC applications, the substitute timeline extends beyond 2030. Indonesia's leverage window is real but finite.

IX. Superforecast to 2030: Three Scenarios

Scenario A — Full Value Chain Achieved (Probability: 20-25%)

Indonesia successfully attracts and retains investment across the full ore → NPI → MHP → pCAM → CAM → cell → pack → vehicle chain. Domestic EV production reaches 500,000+ units by 2030. CATL's Karawang facility operates at 15 GWh, complemented by the HLI Green Power expansion. CNGR's pCAM facility operates at commercial scale. Battery cell exports begin by 2028. The

US-Indonesia critical minerals agreement is upgraded to FTA-equivalent status, unlocking IRA eligibility for at least a portion of Indonesian nickel processed outside FEOC entities.

Key requirements: Stable regulatory environment; successful HPAL technology transfer; decarbonisation investment reducing carbon intensity to EU battery passport compliance thresholds; resolution of FEOC status for non-Chinese-owned Indonesian processing capacity; domestic EV market growth absorbing local production.

Key risks: Policy reversal (Indonesia has reversed resource policy before); sulphur supply disruption (demonstrated in 2026, when Middle East conflict disrupted sulphur supply to HPAL plants, cutting June 2026 MHP output to 29,900 mt from 42,000 mt in January — a 29% decline in six months); environmental regulatory pressure from EU battery passport; nickel price depression from oversupply (2026 surplus estimated at 288,000 tonnes).

Scenario B — Battery Exporter, EV Assembly Fails (Probability: 50-55%)

Indonesia becomes a major exporter of processed nickel (MHP, nickel sulphate) and battery cells — but domestic EV assembly fails to achieve competitive scale. Domestic brands remain marginal. Foreign OEM production (BYD, Hyundai) serves the domestic market adequately but does not generate the local supply chain depth or export volumes that the government targets. Indonesia is plugged into the global battery supply chain as a processing and cell manufacturing hub, capturing significant value, but the vision of a self-sustaining domestic EV industry — building Indonesian brands for Indonesian and export markets — does not materialise.

This scenario reflects the baseline trajectory given the current evidence: Chinese investment in processing and cells is robust; domestic EV demand is growing but slowly; domestic brands face fierce price competition; the incentive structure transition away from import support and toward local content is correct in direction but may not be sufficient in magnitude.

Scenario C — HPAL Technology Problems Limit Processing Growth (Probability: 20-25%)

A persistent sulphur supply crisis (demonstrated in mid-2026), combined with acid tailings environmental liabilities, forces regulatory action or operational shutdowns at multiple HPAL facilities. Several projects miss commissioning timelines. Indonesia's MHP capacity growth undershoots the most optimistic projections (the 2026 range is 694,000 to 922,000 tonnes — a 33% variance that itself reflects significant uncertainty). Nickel prices recover toward \$18,000-\$20,000/tonne as supply falls short of demand. Western HPAL projects (Canada, Australia) become commercially viable at higher price floors. Indonesia retains dominance but the pace of value chain construction slows, and the 2030 battery manufacturing target of 140 GWh is deferred by three to five years.

Trigger signals to monitor: Frequency of tailings failures and regulatory shutdowns; sulphur import volumes (75-80% sourced from Middle East); HPAL utilisation rates (already declining 15-20% in 2026 before the sulphur crisis); EU battery passport enforcement timelines; nickel price trajectory relative to the \$18,000-\$20,000 "sustainable band" Indonesia is targeting.

X. Data Sources Register

See preamble of this document. All market figures sourced from live web searches on 2026-08-28 using Anthropic Claude Code with WebSearch tool. BLMIM API (localhost:8742) was not reachable at time of report generation (connection refused). No training-data market figures have been used in this report. Where ranges are cited, multiple live sources are referenced.

Key data points verified from live sources:

| Data Point | Figure | Source |

| --- | --- | --- |

| Indonesia nickel reserves | ~55 million tonnes / ~42% global share | USGS via Statista, 2023 |

| Indonesia production share 2023 | 54% / 2.02 million tonnes | Mining Technology, ING Think |

| Indonesia production share 2026 | ~50-60% (IEA) | IEA Global Critical Minerals Outlook 2026 |

Indonesia MHP capacity 2026 (range)	694,000 – 922,000 t/yr Ni content	Argus Media, Mining Technology
Indonesia MHP output decline mid-2026	42,000 mt (Jan) → 29,900 mt (Jun)	BC Insight / Bloomberg Technoz
Nickel market surplus 2026	~288,000 tonnes	ING Think
Nickel price range Jan 2026	\$16,500 – \$18,500 / tonne	Trading Economics, Oregon Group
CATL Indonesia investment	~US\$6 billion	CATL official press release
CATL Karawang cell capacity	6.9 GWh initial → 15 GWh	ChinaEVHome
LG/Hyundai HLI Green Power capacity	10 GWh operational + 20 GWh planned	Hyundai Newsroom, EV Magazine
BYD Indonesia factory investment	IDR 11.7 trillion / ~US\$721 million	CnEVPost
BYD Indonesia capacity	150,000 vehicles/year	CnEVPost
Indonesia EV battery target 2030	140 GWh	CSIS
Indonesia EV deployment target 2030	2M cars + 12M motorcycles	ICCT
Indonesia VAT for local EV (2026)	1% (vs 11% standard)	Autoini
IMIP tailings volume	11.5 million tonnes/year	Business & Human Rights Resource Centre
Nickel carbon intensity (Indonesia avg)	93 tCO₂e / tonne Ni	WRI
Indonesia nickel sector share of national emissions	~22% (2023)	WRI
Ford/Vale/Huayou Pomalaa project value	~US\$4.5 billion	Deccan Herald
CNGR total planned investment Indonesia	Up to US\$10.5 billion (3 stages)	The Jakarta Post
Indonesia 2026 nickel ore quota	~250-260 million tonnes	Argus Media, IDNFinancials
US-Indonesia tariff reduction (Feb 2026)	32% → 19%	Indonesia Business Post
ESG investor pressure	US\$4.5 trillion AUM signatories	SBM ITB

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